CARSON SLATER

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SUMMARY

Statistical Science Ph.D. candidate and former collegiate baseball player with a passion for sabermetrics. Researcher and consultant who helps stakeholders turn abstract questions into data-driven decisions and frameworks, with experience handling large insurance and utility data.

SKILLS

General Tools	Positron (≈ VS Code), RStudio, Docker, Git/Github, reveal.js, SQL, duckDB, LaTeX
Python	statsmodels, scikit-learn, polars, pandas, numpy, PyTorch, plotnine, seaborn, plotly
R	tidyverse (dplyr, ggplot2, etc.), TensorFlow (Keras), torch, shiny, RMarkdown/Quarto, rstan, rjags, caret, tidymodels, targets (gittargets, proffer, etc.)
Stats & ML	statistical modeling, hypothesis testing, mixture models for clustering, time series, linear mixed models, Bayesian methods, spatiotemporal modeling, survival analysis, deep learning (convolutional neural networks, denoising diffusion probabilistic models), PCA, tree-based models, SVM
Coursework	Statistical Bioinformatics, Deep Learning, Bayesian Methods/Theory, Survival Analysis, Time Series, Multivariate Statistics, Theory of Statistics, Statistical Methods, Computational Statistics, Numerical Analysis, Machine Learning, Probability Theory, Real Analysis, Intensive Introduction to Computer Science (Harvard), Econometrics

EXPERIENCE

Data Analyst August 2026 - Present
Texas Farm Bureau Insurance Companies, *Part Time* *Waco, TX*

- Built and assessed GLMs as pricing models on 100+ GB datasets to improve the auto insurance rating plan, working within regulatory requirements of the Texas Department of Insurance.
- Ingesting and evaluating new vendor data to assess if purchasing for auto insurance rating would be profitable, evaluating over 200 potential new rating variables.
- Working with millions of policy rows to answer ad-hoc business questions for stakeholders.

Research Assistant January 2024 - July 2026
Baylor University – Research funded by Denver Water *Waco, TX*

- Developed unsupervised learning methods for assessing urban water consumption with IoT meters.
- Extracted, transformed, and loaded unstructured data for analyses for Denver Water collaborators.
- Collaborated with domain experts in urban water demand planning to assess water conservation interventions, implementing rigorous hypothesis testing and building Shiny dashboards to visualize individual and aggregate customer trends.
- Engineered features from household-level billing and water usage time series to perform market segmentation for Denver Water's single-family residential customer base, coming up with seven user personas to characterize their population (∼200k homes) to help drive targeted demand management strategies.
- Edited and reviewed material for an undergraduate intro to data science course.

Statistical Consultant August 2024 - July 2026
Baylor University Statistical Consulting Center *Waco, TX*

- Created and assessed a quantitative framework for assessing the success of college athletics coaches for the Baylor Athletic director. Used selemium to web scrap and compile a large dataset for modeling collegiate coach success using both frequentist and Bayesian mixed effects models.
- Consulted for ongoing interdisciplinary research in the computer engineering and environmental science departments, and for the American Holistic Nursing Association.
- Delivered quality reports for each client and had regular meetings to discuss the project, the methodology and results.

Academic Coach January - May 2023
Wheaton Academy Center for Lifelong Learning *West Chicago, IL*

- Mentored students 1-on-1 to sharpen math skills from Algebra I to AP Statistics.

Mathematics Teaching Assistant August 2022 - May 2023
Wheaton College *Wheaton, IL*

- Facilitated help sessions for undergraduate students in the topics of Machine Learning and Probability Theory.
- Graded homeworks for the teacher of record for Probability Theory and Machine Learning.

Biostatistics Fellow June - July 2022
Duke/North Carolina State Summer Institute for Biostatistics *Raleigh/Durham, NC*

- Engaged in undergraduate discussions and biostatistical methods, clinical trials, and statistical programming.
- Participated in a data hackathon studying myocardial infarction to uncover risk factors for a particular population.

EDUCATION

Ph.D. Statistical Science, Baylor University 2024 - 2027 [Expected]
Advised by Amanda S. Hering · Concentrations in *Data Science* and *Biostatistics*
Research involves developing unsupervised learning methods for time series with applications in municipal water distribution.

M.S. Statistical Science, Baylor University 2023 - 2024

B.S. Applied Mathematics with Statistics, Wheaton College (IL) 2019 - 2023

B.A. Economics
Edman Presidential Scholarship
Varsity Baseball Athlete (NCAA-DIII)

LEADERSHIP EXPERIENCE

President May 2026 - Present
Baylor University Statistical Science Paper Club *Waco, TX*

- Sourcing and facilitating discussions on frontier papers in statistics, biostatistics, and machine learning for graduate students in the statistical science department.

President March 2024 - May 2025
Baylor University Sports Analytics Club *Waco, TX*

- Led a team of undergraduates in submitting a March Madness bracket challenge submission on Kaggle.
- Taught undergraduates basic version control, R programming, statistical and machine learning topics.

PEER-REVIEWED PUBLICATIONS

Brittany J. Carnathan, Dinny Stevens, Swarna Shikha, **Carson Slater**, Nathen Byford, Rodney X. Sturdivant, Kuzy Zarzosa, W. Evan Braswell, and Christie M. Sayes. 2025. "Assessing the Effects of Surface-Stabilized Zero-Valent Iron Nanoparticles on Diverse Bacteria Species Using Complementary Statistical Models". *Journal of Functional Biomaterials* 16 (3). DOI: 10.3390/jfb16030113

MANUSCRIPTS IN PROGRESS

Carson Slater, Ryan Shepler, Samantha Bryan, Tzahi Y. Cath, and Amanda S. Hering. 2026. "Identifying Outdoor Watering Events with High-Frequency Metering Data to Assess Residential Water Consumption". *In Preparation for Submission to Discover Water*.

CONFERENCE PROCEEDINGS & OTHER WORKS

Carson Slater, Ryan Shepler, Samantha Bryan, Tzahi Y. Cath, and Amanda S. Hering. 2026. "Identifying Outdoor Watering Events with High-Frequency Metering Data to Assess Residential Water Consumption". *The 2026 Joint Statistical Meetings*, Thomas M. Menino Convention & Exhibition Center, Boston, MA, August 1-6, 2026. [Poster Presentation]

Jonathan E. Swindell, **Carson Slater**, Samuel Hussey, Charles Baylis, and Robert J. Marks. 2024. "Assessing Interference with Regression Analysis Techniques". *2024 IEEE Texas Symposium on Wireless and Microwave Circuits and Systems (WMCS)*. DOI: 10.1109/WMCS62019.2024.10619025.

Amanda S. Hering, Douglas Nychka, Luke Durell, **Carson Slater**, Michael Poor, Greg Hamerly, and Tzahi Y. Cath. 2024. "Mo(Wa)^{2}TER Introduction to Data Science Course Material". *Harvard Dataverse*. DOI: 10.7910/DVN/BZJW2E.

Carson Slater and Ying Li. 2022. "Forecasting Fast-Moving Consumer Goods with Generalized Additive Models". *2022 University of Illinois Chicago Undergraduate Mathematics Symposium*, University of Illinois - Chicago, Chicago, IL, USA, November 12, 2022. [Poster Presentation]